UNIVERSITEIT VAN PRETORIA
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YUNIBESITHI YA PRETORIA
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Electricity and Electronics: EBN 111
Pre-Practical Assignment 1

Name & Surname:

Student Number:

Instructions:
1. Print the pre-practical assignment.
2. Fill in your student details.
3. Complete all questions found in this assignment.
4. Attach printed graphs to the report.

Question 1

Redraw the circuit shown in Figure 1 on the Orcad Capture CIS program and simulate the circuit. (You can either use a Bias Point Simulation or a Time Domain Simulation; whichever you find more convenient.) Answer the questions that follow.

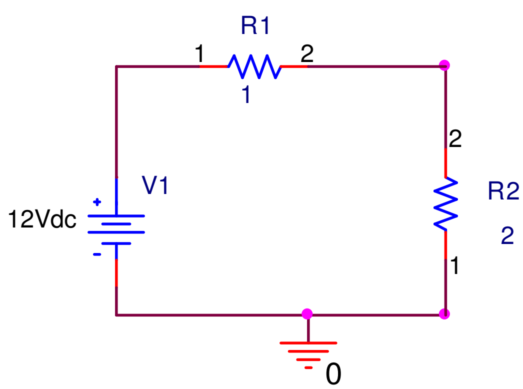

Figure 1

What is the total current in the circuit? **A**

What is the voltage at point 1 of R1? **V**

What is the voltage at point 2 of R1? **V**

What is the voltage difference across R1? **V**

Question 2

Redraw the circuit shown in Figure 2 on the Orcad Capture CIS program.

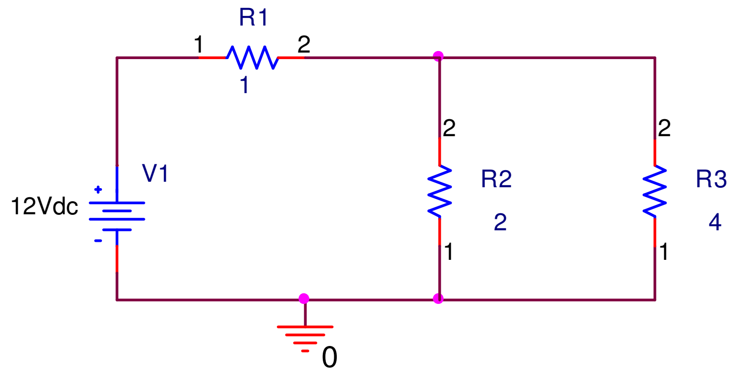

Figure 2

2.1 Complete a Bias Point or Time Domain simulation for Figure 2 and answer these questions.

What is the total current in the circuit?		**A**
What is the current at point 2 of R2?		**A**
What is the current at point 2 of R3?		**A**

2.2 Complete a DC Sweep simulation for Figure 2 sweeping the voltage source V1 from 0V to 12V with steps of 1V.

1. Attach a single graph of a **DC Sweep of V1 from 0 to 12 V** measuring the **current through R1, current through R2 and current through R3.**
2. Attach a single graph of a **DC Sweep of V1 from 0 to 12 V** measuring the **voltage across R1, the voltage across R2 and the voltage across R3.**

(**Note:** Each graph should have three separate lines, one line for each measurement. For each measurement probe that is added in OrCAD Capture, a new line is drawn in the SCHEMATIC window. Each line in the SCHEMATIC window will have the same colour as the colour of the probe in OrCAD Capture. The legend for each line is shown on the left, just below the x-axis of the graph.)